

Emergent Intelligence in Decentralized Swarms

Reward topology, partial observability, and the transition from independent behavior to coordination

Tier 1 - Capstone Research Concept, revised against implemented Stage One

Decentralized Swarm Emergence Research Program

Stage One revision - September 2026

Evidence status: Stage One validates the experimental environment, instrumentation, reward accounting, heuristic baselines, and reproducibility pipeline. It does not yet establish learned emergence, a critical alpha threshold, learned communication, or a phase transition.

Abstract

Decentralized multi-agent systems must transform local observations and local actions into useful collective behavior without relying on a central controller. This project studies whether reward topology can create sufficient alignment for coordinated behavior to emerge under anonymous, radius-limited observations. The implemented Stage One laboratory establishes a reproducible PettingZoo-compatible experimental substrate and frozen heuristic baselines before learned policies are introduced. The central capstone experiment will compare shared-policy PPO and Independent PPO across a pre-registered sweep of the reward-mixing parameter α . Emergence will not be inferred from reward alone: it requires improved global task performance together with structured behavior that is not directly encoded by a controller or heuristic. Stage One currently supports the environment and measurement claims only; the α -dependent emergence hypothesis remains untested until learned-policy data are collected.

1. Research question

Under what reward-topology and observability conditions can decentralized learned agents outperform independent behavior and produce measurable coordination without explicit communication?

2. Scientific motivation

The original capstone framed reward structure as a possible implicit coordination channel. That framing is retained, but the revised study separates infrastructure validation from evidence of emergence. Reward sharing can alter incentives without transmitting explicit messages; whether this produces learned coordination is an empirical question rather than an assumed outcome.

3. Implemented Stage One

- Discrete 16 x 16 world by default, with configurable swarm size; the reference configuration uses six agents and eight targets.
- Anonymous fixed-shape local observations: ego-centered occupancy channels for self, nearby agents, and nearby targets; no agent identifier or absolute position.
- Five simultaneous discrete actions: north, south, west, east, and stay.
- Global weak, global sparse, and hybrid reward accounting with literal per-agent local collection credit.
- Random, greedy, flocking, and distributed-greedy-oracle baselines.
- Episode telemetry for success, completion, movement, collisions, entropy, spatial order, inter-agent distance, role differentiation, efficiency, and return.
- Seeded evidence collection, CSV summaries, manifests, charts, trajectories, and heatmaps.

4. Frozen baseline evidence

Policy	Success	Completion	Mean steps	Mean distance
Random	10%	70%	174.8	787.7
Greedy	100%	100%	13.5	81.0
Flocking	100%	100%	16.5	90.9
Distributed-greedy oracle	100%	100%	34.9	87.4

Across ten independent seeds at $\alpha = 0.5$, greedy completion exceeded random completion in the repository's Welch comparison ($t = -4.609$, $p = 0.00127$). This establishes baseline discrimination, not learned emergence. The 200-condition frozen dataset contains four policies x five alpha settings x ten seeds. Because the heuristic actions are reward-independent, alpha is expected to change recorded return without changing same-seed trajectories.

5. Capstone hypotheses

- H1: Learned coordination performance changes as the global-reward contribution alpha changes.
- H2: Shared-policy PPO and Independent PPO differ in coordination stability under equal training budgets.
- H3: At least one learned condition exceeds stochastic and simple heuristic baselines on task performance while also exhibiting nontrivial coordination structure.
- H4: If a statistically and behaviorally defensible transition exists, an empirical α^* can be estimated; otherwise the null or smooth-response result will be reported.

6. Claim boundary

Current evidence does not establish PPO performance, an alpha-dependent behavioral transition, α^* , learned communication, large-swarm scalability, or resilience to agent failure. The project treats a null result, smooth response, or locally rewarded optimum as scientifically valid outcomes.

7. Capstone finish line

The capstone is complete when shared-policy PPO and Independent PPO have been trained under matched budgets, evaluated on held-out seeded episodes across the alpha sweep, compared with frozen baselines, and analyzed with effect sizes, uncertainty intervals, and behavioral coordination metrics. Communication is intentionally deferred to the next research tier.

Figures

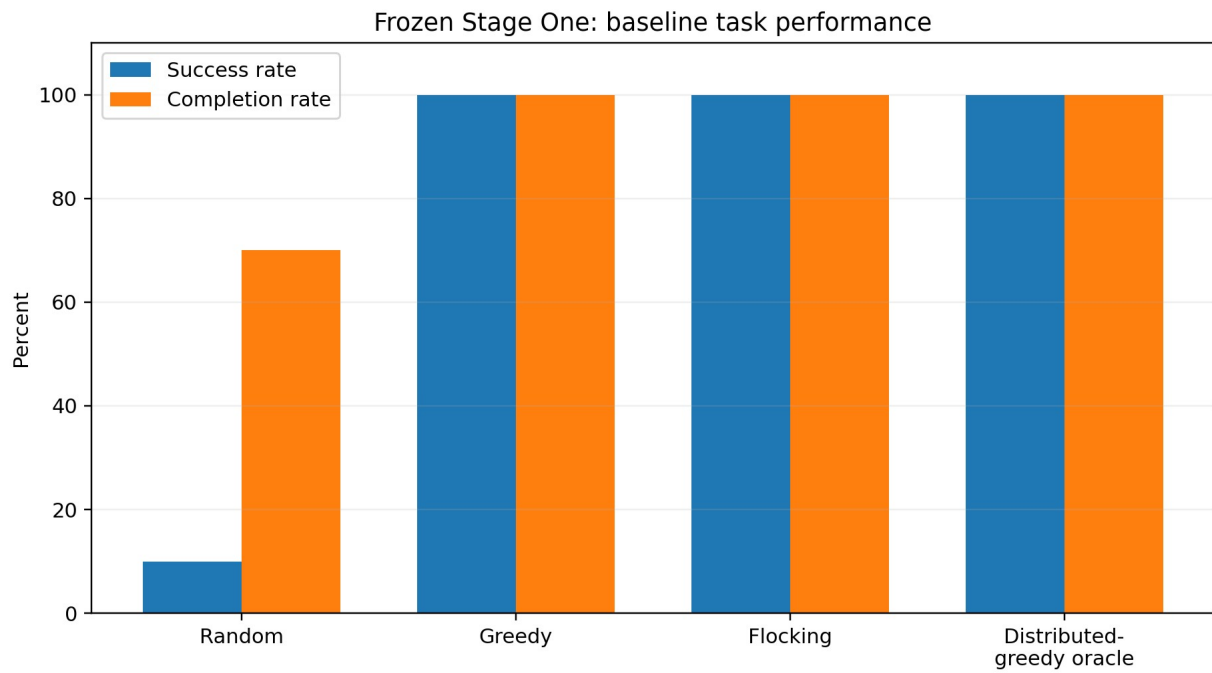


Figure 1. Measured frozen Stage One success and completion rates. These are heuristic/control results, not learned-policy evidence.

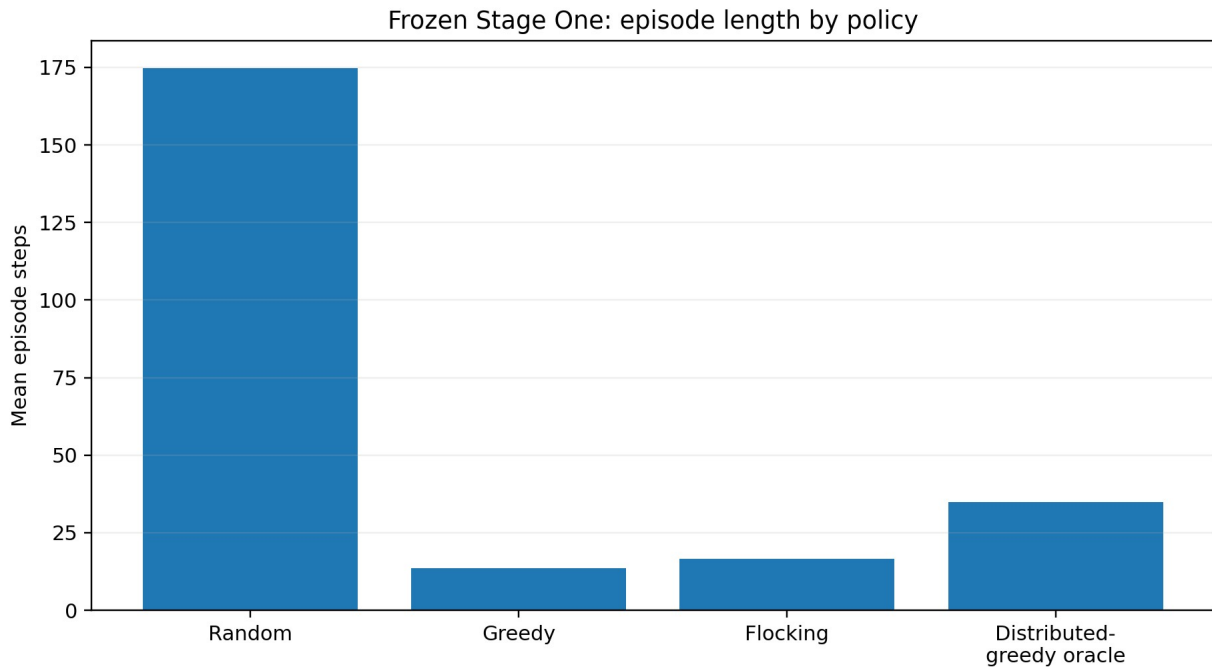


Figure 2. Measured mean episode length for the frozen Stage One baselines.

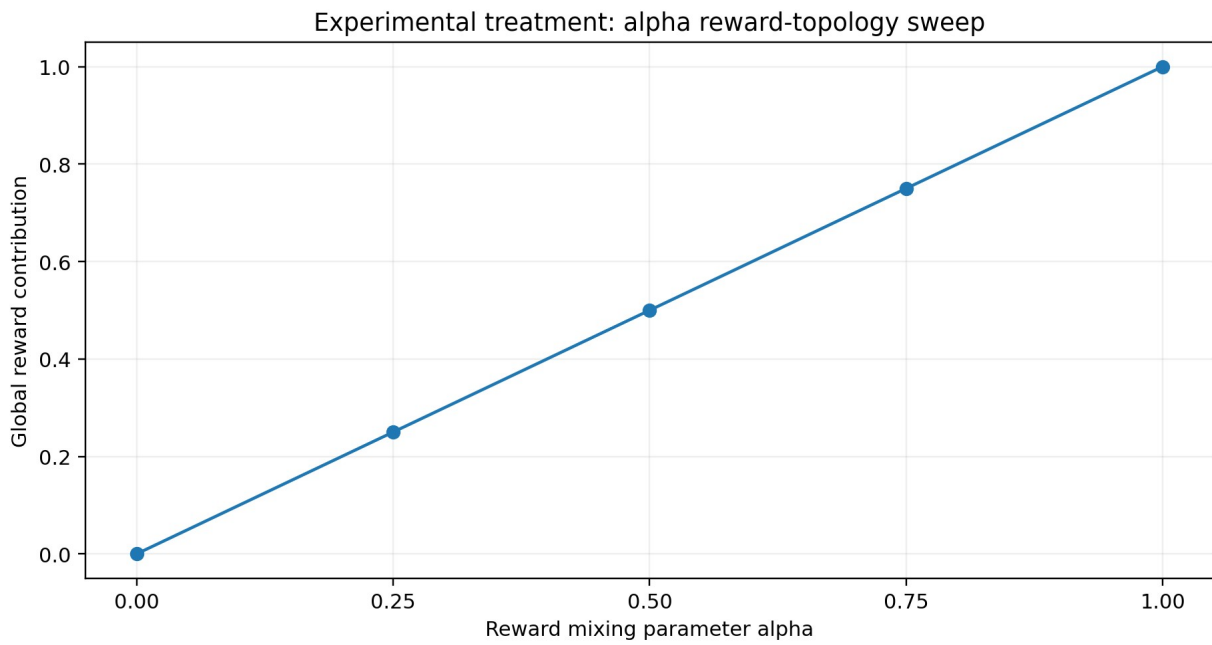


Figure 3. Experimental alpha treatment planned for the learned-policy study. This figure shows the treatment definition, not an observed emergence curve.